

Task 11 — Matching & Ranking v2 (Learning-to-Rank)

Data Analyst · Sprint C - Intelligence Layer · Task Brief

DATA ANALYST

PlaceMux · Phase 3 · Task Brief

TASK 11 / 25

Deep Specialist · Expert

Sprint C - Intelligence Layer · Mastery 77%

Focus

Build the evaluation framework for matching quality: is the marketplace actually matching well?

What to build / complete

- A match-quality metric set (relevance, shortlist rate, hire rate, time-to-hire)
- A labelled evaluation set for offline ranking assessment
- A quality baseline the new ranker must beat

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: You can say objectively whether matching improved, in terms the business actually cares about.

Stage B — Build: A match-quality metric set (relevance, shortlist rate, hire rate, time-to-hire)

1. Define the metric, its source & the decision

Define precisely what “A match-quality metric set (relevance, shortlist rate, hire rate, time-to-hire)” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: A labelled evaluation set for offline ranking assessment

1. Define the metric, its source & the decision

Define precisely what “A labelled evaluation set for offline ranking assessment” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: A quality baseline the new ranker must beat

1. Define the metric, its source & the decision

Define precisely what “A quality baseline the new ranker must beat” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Present the quality baseline and demonstrate how a ranking change would be judged against it.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “A match-quality metric set (relevance, shortlist rate, hire rate, time-to-hire)” is complete, real, and demoable end-to-end.
- “A labelled evaluation set for offline ranking assessment” is complete, real, and demoable end-to-end.
- “A quality baseline the new ranker must beat” is complete, real, and demoable end-to-end.
- Good looks like: You can say objectively whether matching improved, in terms the business actually cares about.

Worry if any of these are true

- Judging matching purely on clicks.
- An evaluation set with unvalidated labels.
- Calling an experiment early because it looked good on day two.

Hand-off

- Quality baseline to AI-ML. Make sure the next person can pick it up without you.